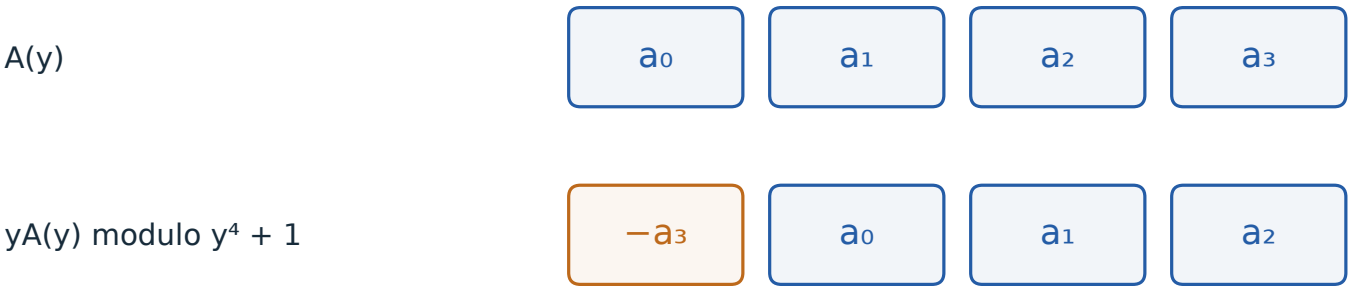

The wrapped coefficient changes sign because $y^4 = -1$.